UNIVERSITY OF LONDON

CM3020

BSc EXAMINATION

COMPUTER SCIENCE

Artificial Intelligence

Release date: Monday 5 September 2022 at 12:00 midday British Summer Time

Submission date: Tuesday 6 September 2022 by 12:00 midday British Summer Time

Time allowed: 24 hours to submit

INSTRUCTIONS TO CANDIDATES:

Section A of this assessment paper consists of a set of **TEN** Multiple Choice Questions (MCQs) which you will take separately from this paper. You should attempt to answer **ALL** the questions in Section A. The maximum mark for Section A is **40**.

Section A will be completed online on the VLE. You may choose to access the MCQs at any time following the release of the paper, but once you have accessed the MCQs you must submit your answers before the deadline or within **4 hours** of starting whichever occurs first.

Section B of this assessment paper is an online assessment to be completed within the same 24-hour window as Section A. We anticipate that approximately **1 hour** is sufficient for you to answer Section B. Candidates must answer **TWO** out of the THREE questions in Section B. The maximum mark for Section B is **60**.

Calculators are not permitted in this examination. Credit will only be given if all workings are shown.

You should complete **Section B** of this paper and submit your answers as **one document**, if possible, in Microsoft Word or a PDF to the appropriate area on the VLE. Each file uploaded must be accompanied by a coversheet containing your **candidate number**. In addition, your answers must have your candidate number written clearly at the top of the page before you upload your work. Do not write your name anywhere in your answers.

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SECTION A

Candidates should answer the **TEN** Multiple Choice Questions (MCQs) quiz, **Question 1** in Section A on the VLE.

SECTION B

Candidates should answer any **TWO** questions from Section B.

Question 2

Markov Decision Process

Consider the problem of creating a control system for a self-driving car. For our purposes, we shall consider a simple self-driving car which is just driving around the street without needing to go anywhere in particular (i.e. no navigation). The objective is for the car to avoid crashing and to keep moving. The car is completely autonomous and it has various sensors and motors to enable it to perceive the world, to drive and to steer. We are going to attempt to model this problem using a Markov Decision Process.

(a) Name THREE elements of a Markov Decision Process and explain what they would represent in the self-driving car control system problem. [9]

(b) You decide to use Q-learning to help with the solving of the Markov Decision Process for the self driving car. You decide to use a neural network to represent the Q-function. Explain what the Q-function does in the context of the self-driving car control system and how it relates to the Markov Decision Process. [3]

(c) What would be the training data for the neural network? Explain what the elements of the training data represent in relation to the self-driving car problem. [6]

(d) In the Atari game player example seen in the the AI course, the training data in Q-learning changes over time. Explain why this is the case. [3]

(e) Once the neural network is trained, what would be the inputs and outputs during use? [3]

(f) In order to train and test the self-driving system, it might be necessary to use a simulation. Describe briefly what the simulation might consist of. Explain how the Markov Decision Process and the Q-function would interact with the simulation and how it would be used to train the network. [6]

Question 3

Artificial Intelligence and Creativity

(a) What was the name of the system used in the pop song generator case study from the AI course to generate musical fragments? Unpack the acronym into the words that make it up. [2]

(b) The musical fragment generator you just named has a decoder and an encoder component. Provide text for each of the blanks in the following paragraph:

The encoder takes as its input i data. This data comes in the form of ii which represent iii. The output of the encoder can be described as a iv. This output comes in the form of v. The output of the encoder represents a position in the vi of the model.

i. [1]
ii. [2]
iii. [2]
iv. [2]
v. [2]
vi. [2]

(c) Describe THREE data or software components required to generate a musical fragment from a pre-trained model of the sort used to generate musical fragments in the pop song case study. Use diagrams if they help. [6]

(d) Write a step by step process from a programmer's perspective how you would go about generating a musical fragment from this system, assuming they have already been trained. [3]

(e) According to Boden (Boden, M. A. (2009). Computer models of creativity. AI Magazine, 30(3), 23-23.), what TWO features do the ideas that a system creates need to possess in order to be 'creative'? Explain what those terms mean. [4]

(f) For each of the TWO properties of ideas you mentioned in the previous answer, discuss if the musical fragment generator component of the pop song generator can potentially have that property. Would it be possible to measure that property somehow? [4]

Question 4

Genetic algorithms

(a) Explain the purpose of the following aspects of genetic algorithms:

i. Genetic encoding. [2]

ii. Fitness function. [2]

iii. Genetic manipulations. [2]

(b) Draw a diagram showing the cycle that a genetic algorithm goes through, making it clear which data is being processed at each step and which parts are computationally expensive. [4]

(c) What is the difference between evolving neural network architectures using a genetic algorithm and training a neural network using a typical algorithm such as back-propagation? [4]

(d) In the creatures case study, the lecturer mentioned Inman Harvey who pointed out that a lot of the creativity and design time required to build a genetic algorithm is spent on the encoding scheme. Based on your experience with the evolving creatures, do you agree with this? Justify your answer by comparing the complexity of the genetic encoding scheme to that of the other components in the creatures system. [6]

(e) Select a reference from the set provided with the genetic algorithm case study wherein the researchers evolved something. Provide a citation and describe in your own words what the problem was that the researchers were trying to solve with the genetic algorithm. [4]

(f) A company is proposing to use a flight control system on a passenger aeroplane which they designed using a genetic algorithm. Explain why you think this system is likely to be safe or not, compared to a human-design system. What would you do to improve the chances of producing a safe control system? [6]

END OF PAPER

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